

New Channels & Deep Learning

Broadening LDMX's Dark Matter Search

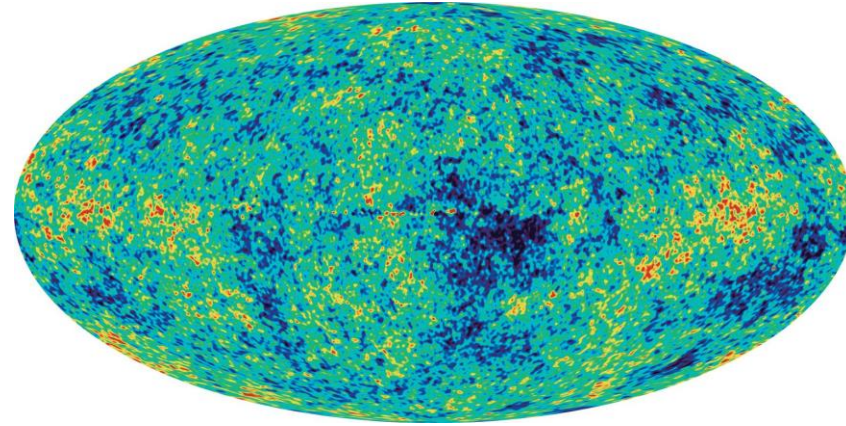
Kieran Wall



We know Dark Matter is out there!

Gravitational Lensing Surveys

arXiv:2601.14559



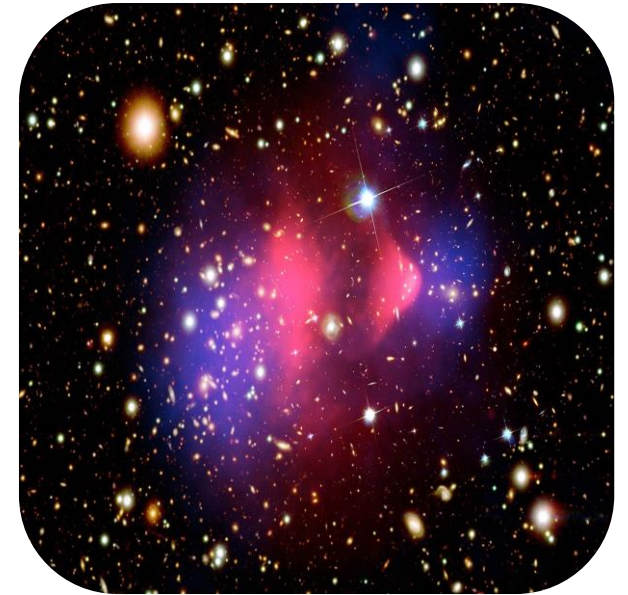
Cosmic Microwave Background Power Spectrum

arXiv:0603.449

But no evidence in the lab ... yet

Mass vs. Gas in Bullet Cluster

arXiv:0608.407



Light Dark Matter

- If DM is a new particle, cosmological evidence does not place strong constraints on its mass
- Candidates span **70** orders of magnitude (!!)
- By assuming DM was in **thermal equilibrium** in early universe → **dramatically reduce range**



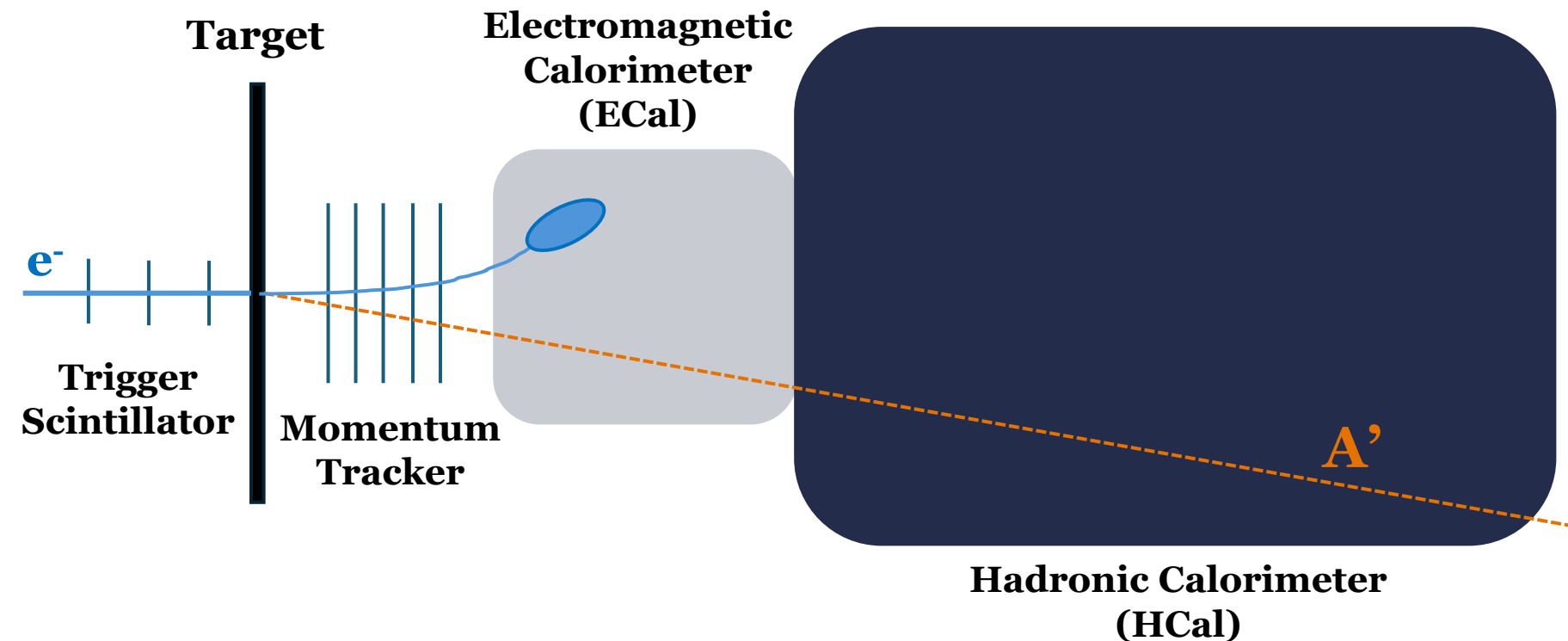
- Open parameter space ☺
- **Focus of LDMX**
- Focus of decades of experiments
- Restricted parameter space (for now) ☹



Will search for light dark matter at SLAC National Accelerator laboratory with a missing momentum approach

LDMX and Missing Momentum

- LDMX is designed to detect the missing energy and momentum signature of a dark photon (A') which in many models **couples light dark matter**
- A process which could allow us to indirectly detect a dark photon is **dark bremsstrahlung**

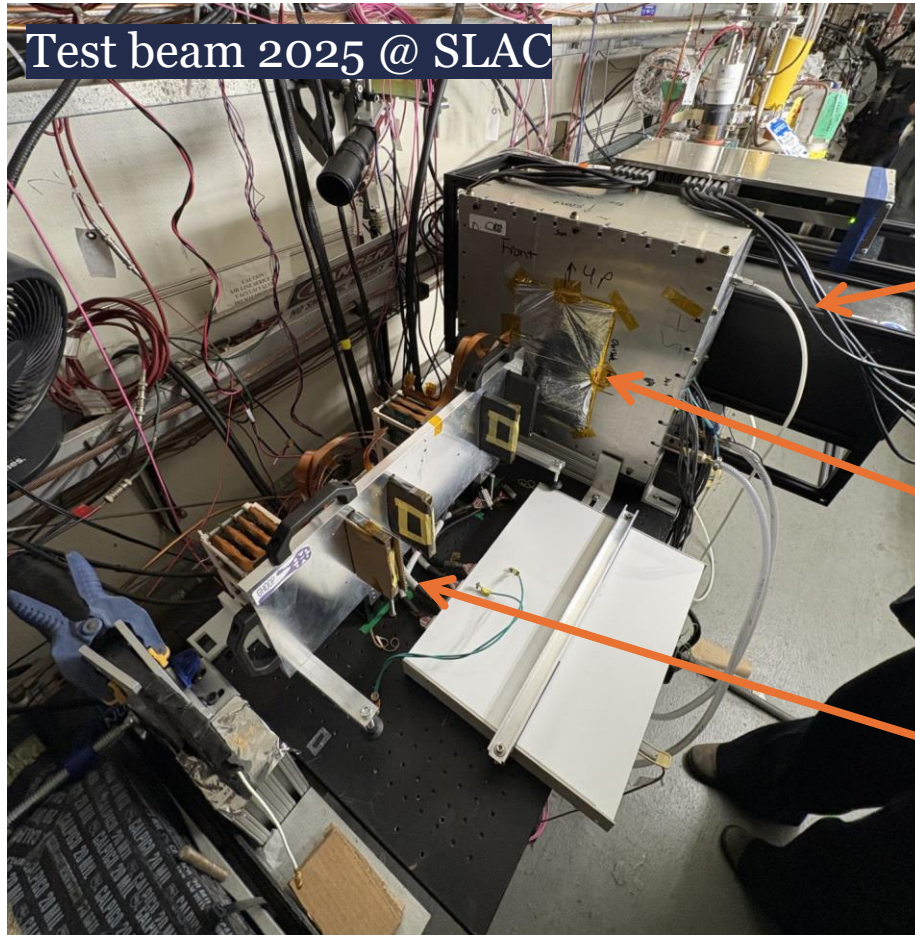


Signal Event Signature:

- Single low energy electron with “missing” energy and momentum, and nothing else in calorimeters

LDMX and Missing Momentum

- Cartoons are great, but components of LDMX are being built and tested today!



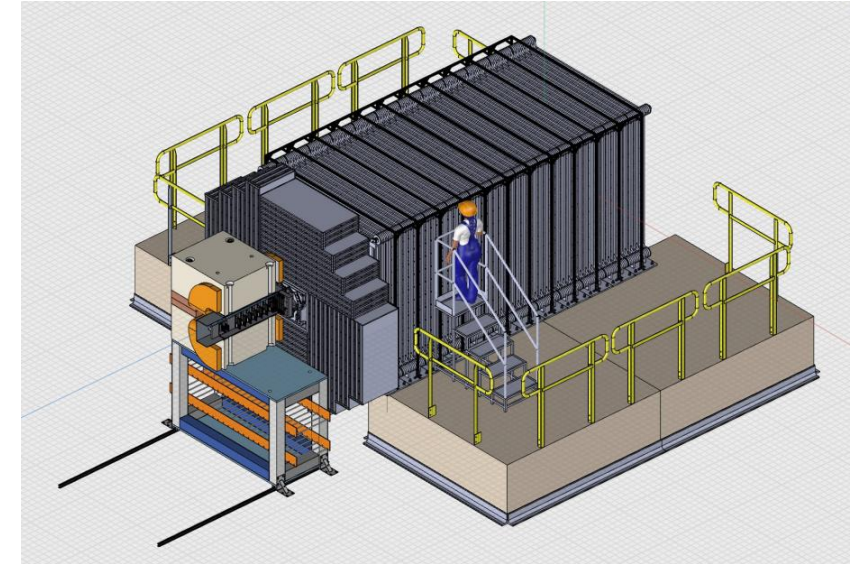
Test beam 2025 @ SLAC

HCal Prototype



ECal Prototype

**Trigger
Scintillator
Prototype**

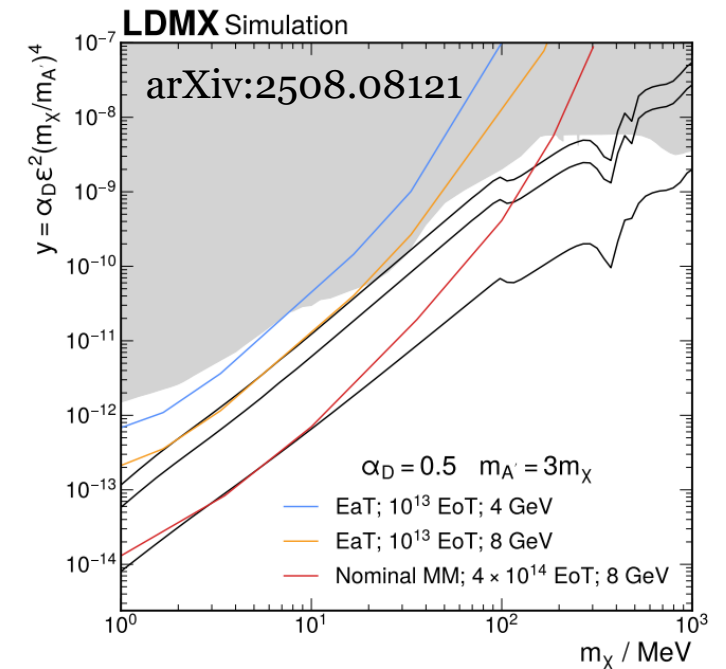
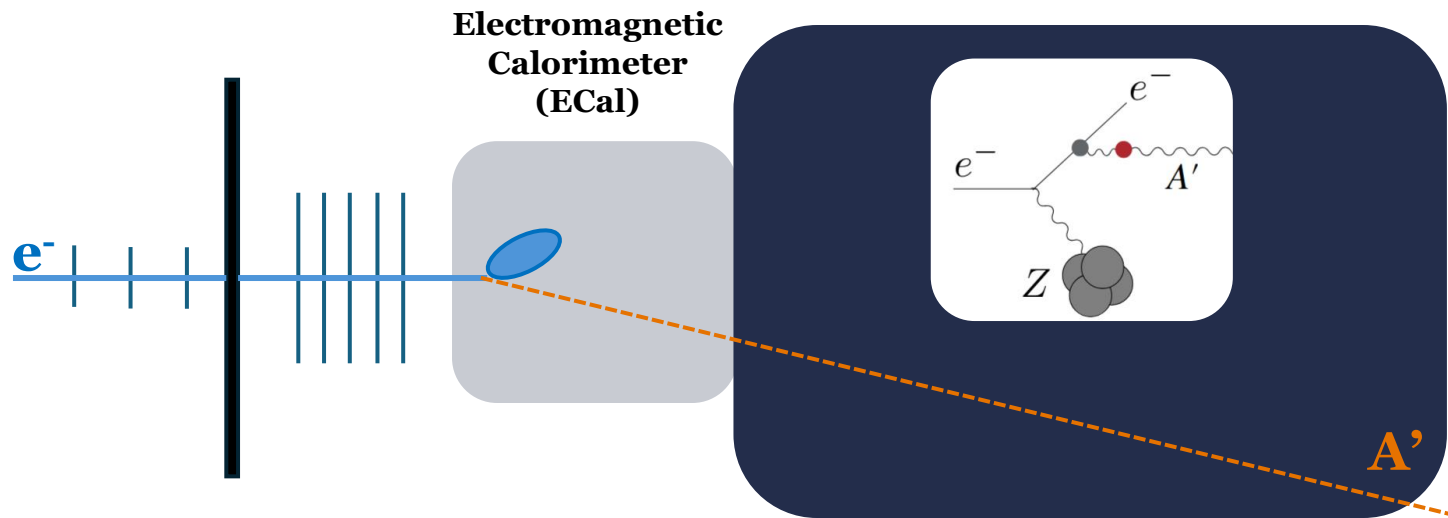


The LDMX Missing Momentum search will cover three orders of magnitude of unexplored parameter space when it runs

But it can explore even more through complementary channels!

ECal-as-Target and Visible A's

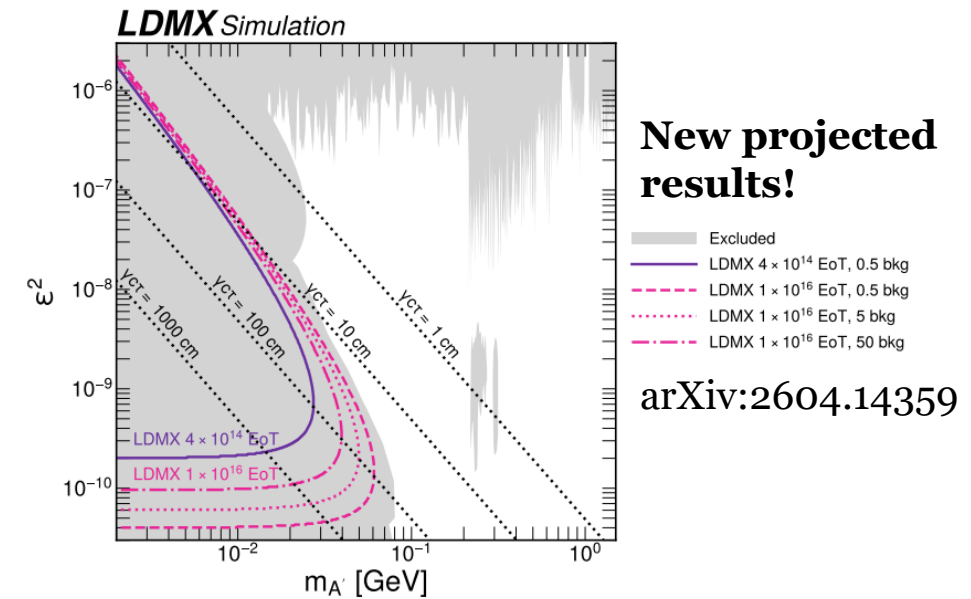
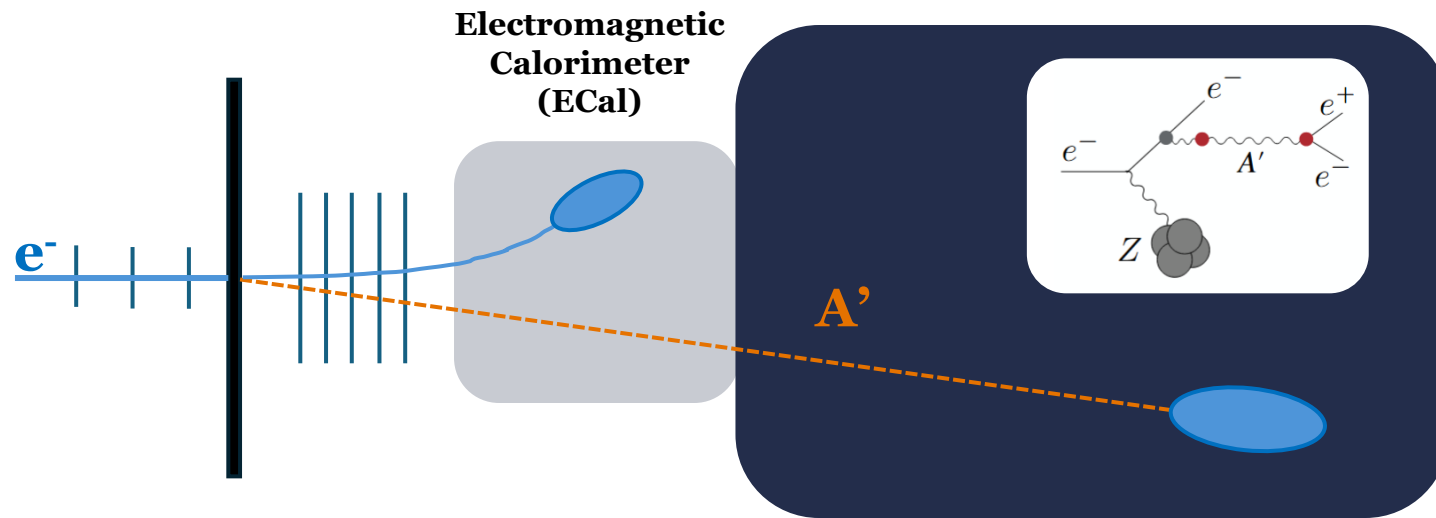
- LDMX can search for DM beyond the missing momentum channel
- Early-run ECal-as-Target (EaT). Look for missing energy in ECal from dark brems at ECal front face



- Allows LDMX to collect electrons-on-target while recoil tracker is still being commissioned and make use of electrons which do not interact in the target.

ECal-as-Target and Visible A's

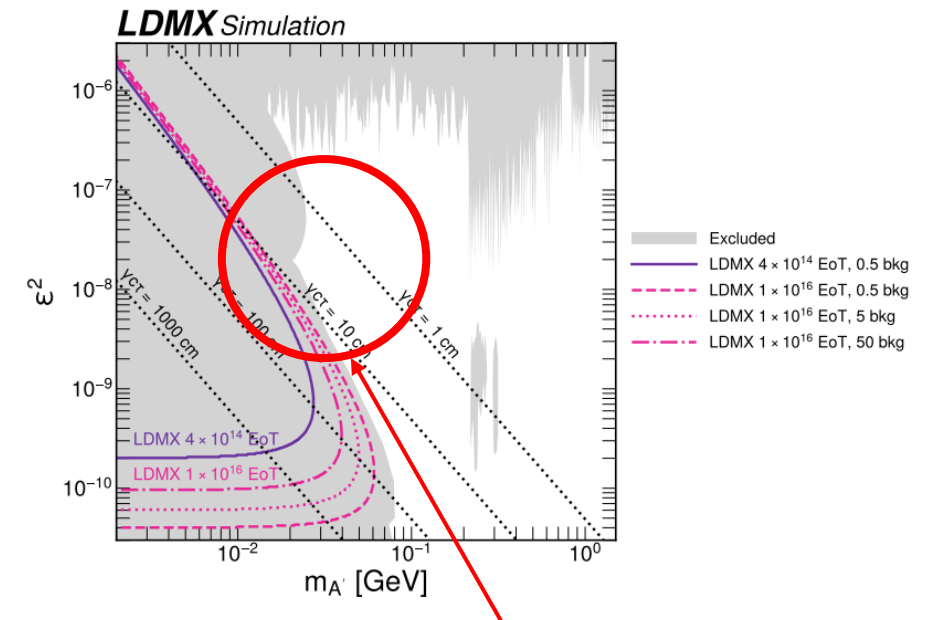
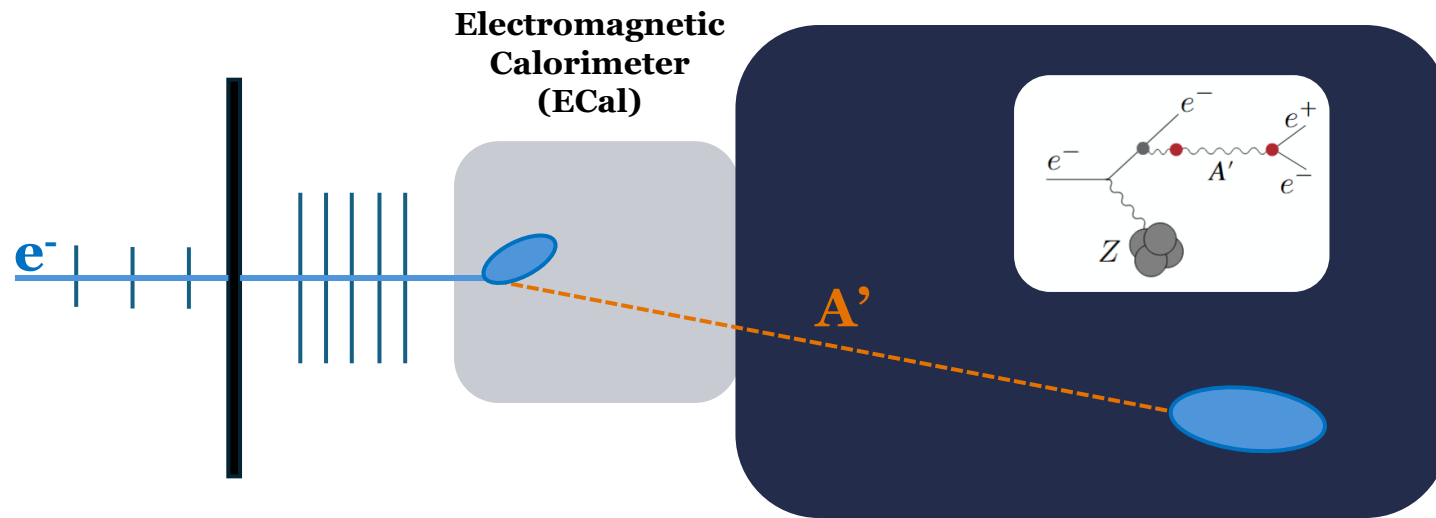
- We can also consider an alternative signal channel, where $m_{A'} > 2m_{\text{DM}} > 2m_e$
- This allows the A' to decay **visibly into an electron-positron pair**. “Long-Lived-Particle” (LLP)
- New signal profile $\rightarrow$ Soft recoil electron in ECal, displaced EM shower in HCal



- EaT boosts luminosity of missing momentum search, LLPs search expands LDMX's reach into entirely new thermal dark matter models.
- **Can you guess what one more channel is??**

EaT Visibles – Combining Alternative Channels

- LLP signal with EaT production location, dark brem at ECal front face, visible decay in HCal.



- Motivation: EaT luminosity boost, probes larger interaction strengths (red region on plot)
- Opportunity to explore new background rejection techniques!

EaT Visibles Selection Cuts

- There are many Standard Model processes that mimic our EaT Visibles signal, and even more than don't look like it we can reject immediately
- Can develop and test selection cuts using large Monte-Carlo simulations of signal and background

Cut	Remaining Bkgd.	Remaining Signal (100 MeV A')
Trigger < 3160 MeV in front ECal	.0005%	93.4%
Total ECal Energy < 3160 MeV	.00011%	88.2%
Total HCal Energy > 4440 MeV	.00004%	68.7%

We can eliminate more background by taking a closer look at the topology of the energy depositions in the ECal and HCal.

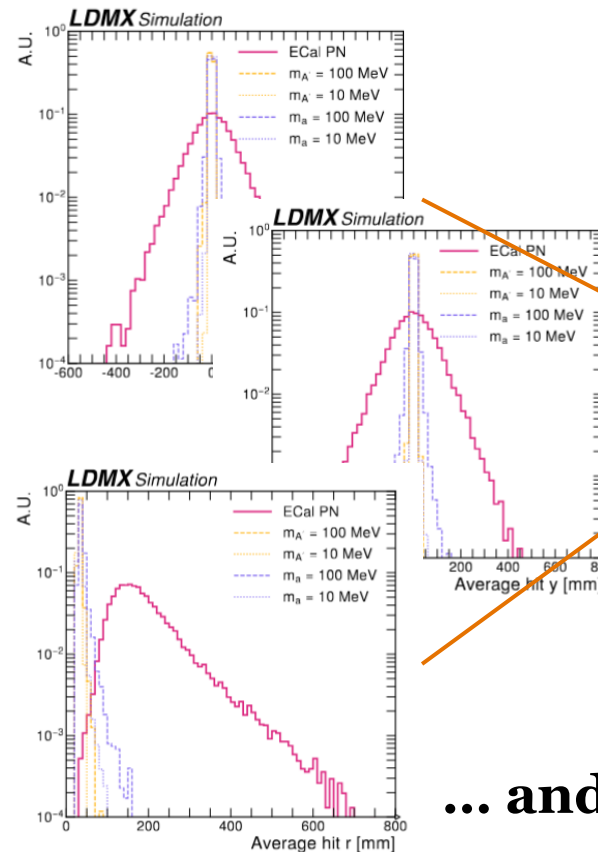
For 1E13 EoT, this is still over 7 million passing background!!

Smart Discriminators – BDT vs GNN

- EaT visibles signals should have distinct topological signatures in the ECal and HCal.
- Machine learning algorithms can identify these features and synthesize them into a single **discriminator score**.

BDT

- Boosted-Decision-Trees utilize gradient boosting to optimally identify signal events based on input features
- 47 features computed from ECal hits, and 11 from HCal hits
- BDTs:
 - Commonly used
 - Interpretable
 - Efficient w.r.t training data.



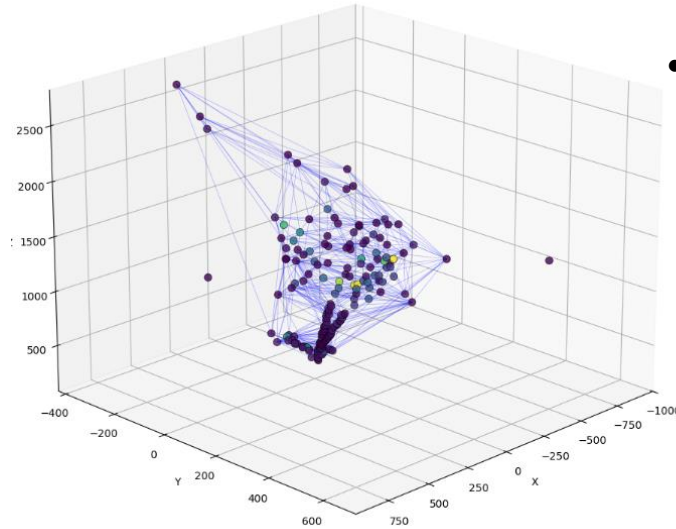
Merged to Single
Discriminator Score

... and 55 more features

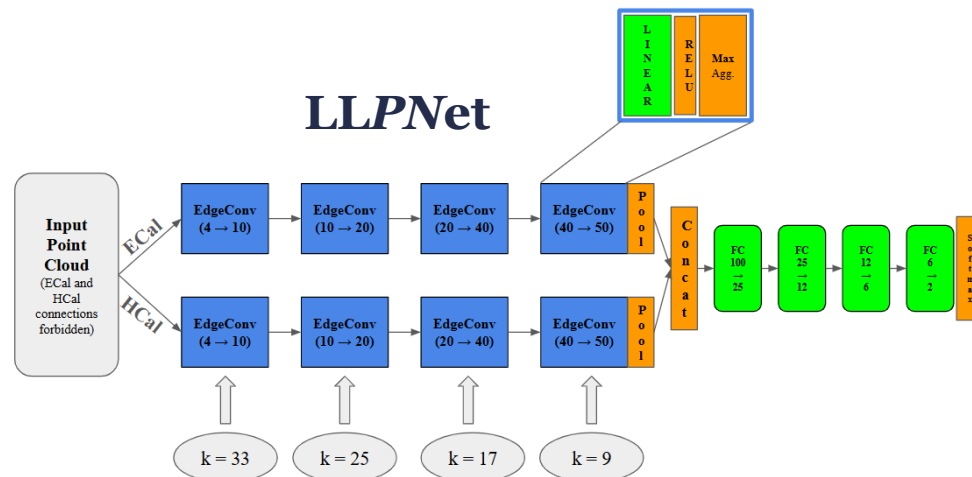
Smart Discriminators – BDT vs GNN

GNN

- Graph-neural-networks are neural networks which work with graphs as input objects
- Our GNN creates point-clouds for the ECal and HCal, utilizes several layers of EdgeConv message passing modules, and builds representations across detectors with an MLP
- Graph generated dynamically with k-nn clustering between layers



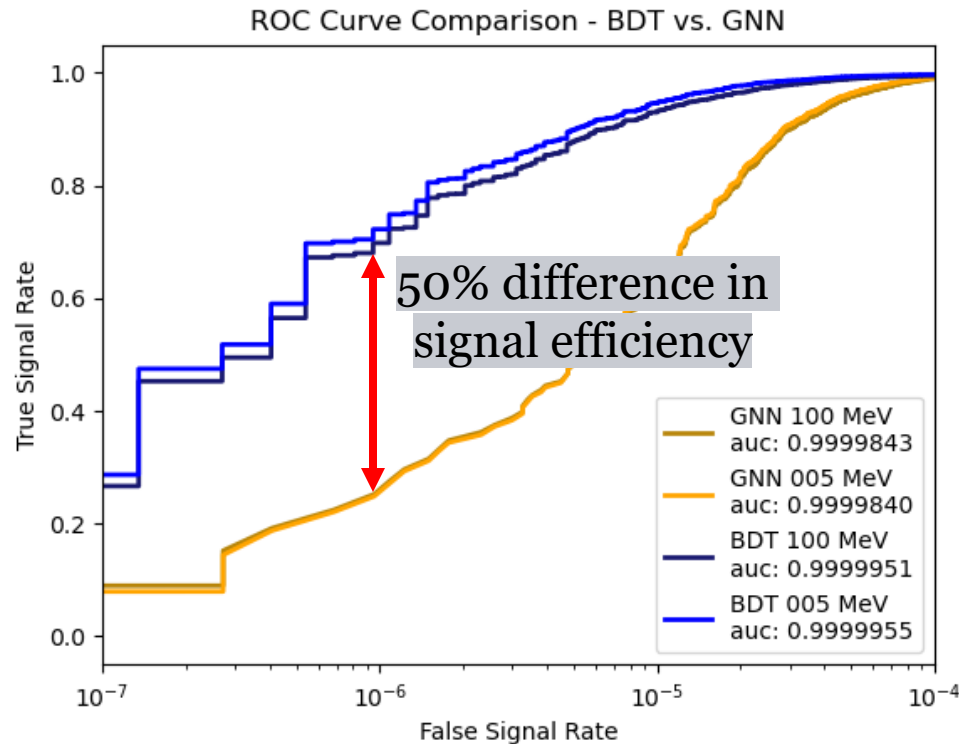
- GNNs:
 - Builds its own features, can learn higher dimensional features.
 - Less interpretable
 - *Increasingly popular in particle physics*
 - arXiv:1902.08570



Two very different approaches to signal vs. background discrimination. **Let's see which performs better**

Discriminator Faceoff – BDT vs. GNN

- BDT and GNN **trained** on *identical* **1e12** EoT samples. **Tested** against a **3e13** EoT sample...



- From ROC curve, we see superior performance from the BDT, higher signal efficiency for all background rejection rates
- BDT can veto **ALL background** at **26.6%** discriminator efficiency → no such GNN threshold
- GNN overconfidence*

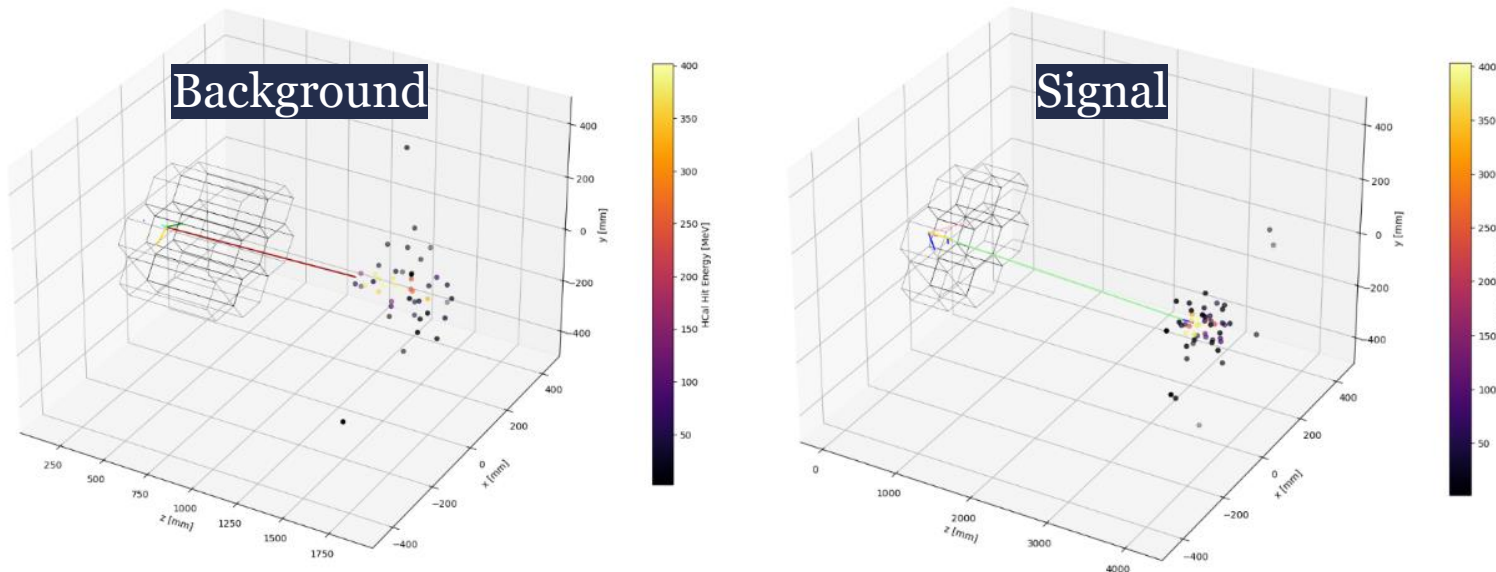
Takeaways

- LDMX Missing Momentum Search will revolutionize our understanding of the light dark matter model
- Alternative search channels can boost luminosity and explore complementary models
- Deep learning is a powerful approach to background rejection
 - Throwing a neural network at a problem is NOT always the solution
 - Good models require thought, experimentation, and persistence!
- Thanks to VAS for giving me a place to talk about my research for the past three years!



Backup: Discriminator Faceoff – BDT vs. GNN

- BDT has considerable head start on GNN, fed 58 features chosen for their ability to distinguish signal and background
- GNN shows signs of being too shallow → additional layers and parameters will increase performance
- The GNN is learning real physics (!!)



Top 10 highest GNN score background events are **kaons**, a long-studied, difficult to veto background in LDMX searches

Likely not overfitting on simulation noise 😊